Paul G. Beckman

Updated August 3, 2026

CV

Education

PhD in Mathematics — *Courant Institute, New York University* — 2020-2025
Advisor: Michael O'Neil
Thesis: Fast transform methods for Gaussian random fields

BS in Computational and Applied Mathematics — *University of Chicago* — 2015-2019
Advisor: Mihai Anitescu
Thesis: Nonstationary Gaussian process approximations of piecewise analytic computer codes

Research Positions

Peter O'Donnell Jr. Postdoctoral Fellow — *Oden Institute, UT Austin* — 2025-present
Advisors: Joe Kileel, Per-Gunnar Martinsson

CSGF Practicum — *Lawrence Berkeley National Laboratory* — 2023
Advisors: Yang Liu, Xiaoye Sherry Li

Mathematics and Statistics Researcher — *Argonne National Laboratory* — 2019-2020
Advisors: Mihai Anitescu, Michael Stein

Publications

David Persson, **Paul G. Beckman**, Tyler Chen, Diana Halikias, Christopher Musco. "A Recursive Butterfly Factorization with Optimality Guarantees." arXiv preprint arXiv:2607.29361 (2026).

Paul G. Beckman, Samuel F. Potter, Michael O'Neil. "A Butterfly-Accelerated Manifold Harmonic Transform." arXiv preprint arXiv:2605.21828 (2026).

Beckman, Paul G., Michael O'Neil. "A Nonuniform Fast Hankel Transform." *SIAM Journal on Scientific Computing*, no. 2 (2026): A784-A803.

Christopher J. Geoga, **Beckman, Paul G.**. "Nonparametric spectral density estimation from irregularly sampled data." arXiv preprint arXiv: 2503.00492 (2025).

Beckman, Paul G., Christopher J. Geoga. "Fast Adaptive Fourier Integration for Spectral Densities of Gaussian Processes." *Statistics and Computing* 34, no. 6 (2024): 217.

Beckman, Paul G., Christopher J. Geoga, Michael L. Stein, and Mihai Anitescu. "Scalable Computations for Nonstationary Gaussian Processes." *Statistics and Computing* 33, no. 4 (2023): 84.

Williams–Young, David B., **Paul G. Beckman**, and Chao Yang. "A Shift Selection Strategy for Parallel Shift-Invert Spectrum Slicing in Symmetric Self-Consistent Eigenvalue Computation." *ACM Transactions on Mathematical Software (TOMS)* 46, no. 4 (2020): 1-31.

Awards

Moses A. Greenfield Research Prize — *Courant Institute* — 2024

Computational Science Graduate Fellowship — *Department of Energy* — 2020

Presentations

CodEx Seminar — *CodEx Seminar* — 2026
"Fast transforms for Gaussian random fields"

UT Austin — *Oden Institute Seminar* — 2026
"Fast transforms for Gaussian random fields"

UT Austin — *Data & Algebra Seminar* — 2026
"Implicit tensor moment estimation of nonparametric mixture models with sparse dependence"

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SIAM *Uncertainty Quantification* 2026
"Fast transforms for spectral modeling of Gaussian processes"

SIAM *Computational Science and Engineering (Minisymposium co-organizer)* 2025
"Fast adaptive Fourier integration of spectral densities"

Princeton University *PACM IDeAS Seminar* 2025
"A nonuniform fast Hankel transform"

SIAM *Uncertainty Quantification* 2024
Talk: "Fast adaptive Fourier integration of spectral densities"
Poster: "Butterfly-accelerated Gaussian random fields on manifolds"

ICIAM *International Congress on Industrial and Applied Mathematics* 2023
"Boundary integral methods for computing covariances in inverse source problems"

SIAM *Mathematics of Data Science (Minisymposium co-organizer)* 2022
"Fast algorithms for elliptic PDEs with Gaussian boundary noise"

Teaching

Differential Calculus	*UT Austin M408K (TPEI)*	Co-instructor	Fall 2026
Mathematic Statistics	*NYU MATH-UA.2340*	Teaching Assistant	Spring 2024
Statistics	*NYU MATH-GA.2962*	Teaching Assistant	Fall 2021
Computational Statistics	*NYU MATH-GA.2080*	Teaching Assistant	Spring 2021

Software

Primary developer `FastHankelTransform.jl`, `SpectralKernels.jl`

Contributor `FastGaussQuadrature.jl`, `chunkIE`

Outreach and Service

Peer reviewer
Journals include *IEEE Signal Processing Letters* and *Statistics and Computing*.

Volunteer Instructor *Texas Prison Education Initiative (TPEI)* 2026-present
Co-instructor for credit-bearing college math courses for incarcerated youth and adults

Volunteer Instructor *Petey Greene Program* 2020-2025
Co-instructor for elementary through college level math for currently and formerly incarcerated adults

Reading Group Founder *Courant Institute* 2020-2024
Co-founded a department group reading articles on progress towards more accessible math higher education

Department Climate Survey Designer *Courant Institute* 2020-2022
Co-designed, administered, and reviewed a survey on Courant Institute PhD student experience

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